Karyotype: 48/R/50,XY,+X,-Y,-3,-6,-7,del(7)(q?31q36),+8,+11,add(12)(p11.2),
+14,+15,-16,+19,+20,+22- ,+1/R/2mar[cp10] -Amendment 14/10/20

FISH Result: 7q deletion detected. -Amendment 14/10/20

-Amendment 14/10/20 Chromosome analysis has shown all 10 cells examined to have abnormal hyperdiploid karyotype with abnormalities which included: a deletion of the long arm of chromosome 7; additional material on the short arm of chromosome 12 (resulting in loss of 12p); an extra copy each of chromosomes X, 8, 11, 14, 15, 19, 20 and 22; loss of one copy each of chromosomes Y, 3, 6, 7 and 16; the presence of up to 2 unidentified marker chromosomes. However, chromosome morphology was poor and a small structural rearrangement may not have been excluded.

Fluorescence in situ hybridisation (FISH) studies were carried out using the Cytocell MyProbe tricolour gene probes [Del(7q) Plus] to detect deletions of chromosome 7 at 7q36.1 and 7q22.1 and monosomy 7 seen in myeloid neoplasia, including MDS and AML.

93 out of 100 interphase cells examined showed loss of the 7q36.1 gene probe confirming deletion of 7q seen on routine analysis.

Hyperdiploidy, gains of chromosomes 8 and 11, and loss of 7q and 12p are recognised findings in myeloid neoplasms including AML. In particular a complex karyotype is associated with an adverse prognosis in AML (Grimwade D et al, Blood. 2010 Jul 22;116(3):354-65). -Amendment 14/10/20

Conclusion: Complex abnormal hyperdiploid karyotype with gains of chromosomes 8 and 11, and loss of 7q and 12p which are recognised findings in myeloid neoplasms including AML - Amendment 14/10/20